

Lecture Series 6: Statistics of Agreement CM2018

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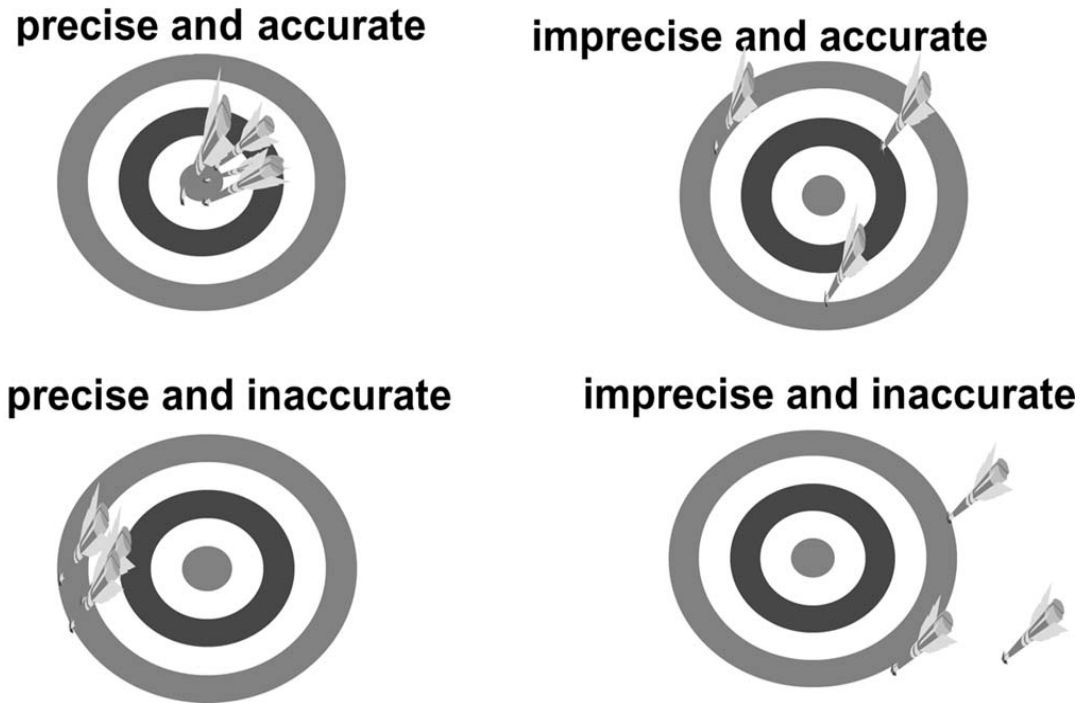
L6-1 Statistics of Agreement (I)

Statistics of agreement

- Do different measurements give the same result?
- How close are two measurements?
- Comparison between measurements with the same method.
- Comparison between different measurement methods.
- Comparisons of measurements against a reference method, gold standard or ground truth (≈ correctness).

Accuracy vs Precision





- Agreement is characterised by the absence of both *systematic* and *random errors*.
- **Accuracy** absence of systematic errors (bias)
- **Precision** absence of random errors (noise)

(Lin 2008)

Types of Precision

- **Repeatability** precision under ”tight conditions”. Same equipment, different days or different batches/measurement series
- **Reproducibility** precision across labs or devices
- **Reliability** variation across measurements relating to the variation in the population

Accuracy of Continuous Measurements

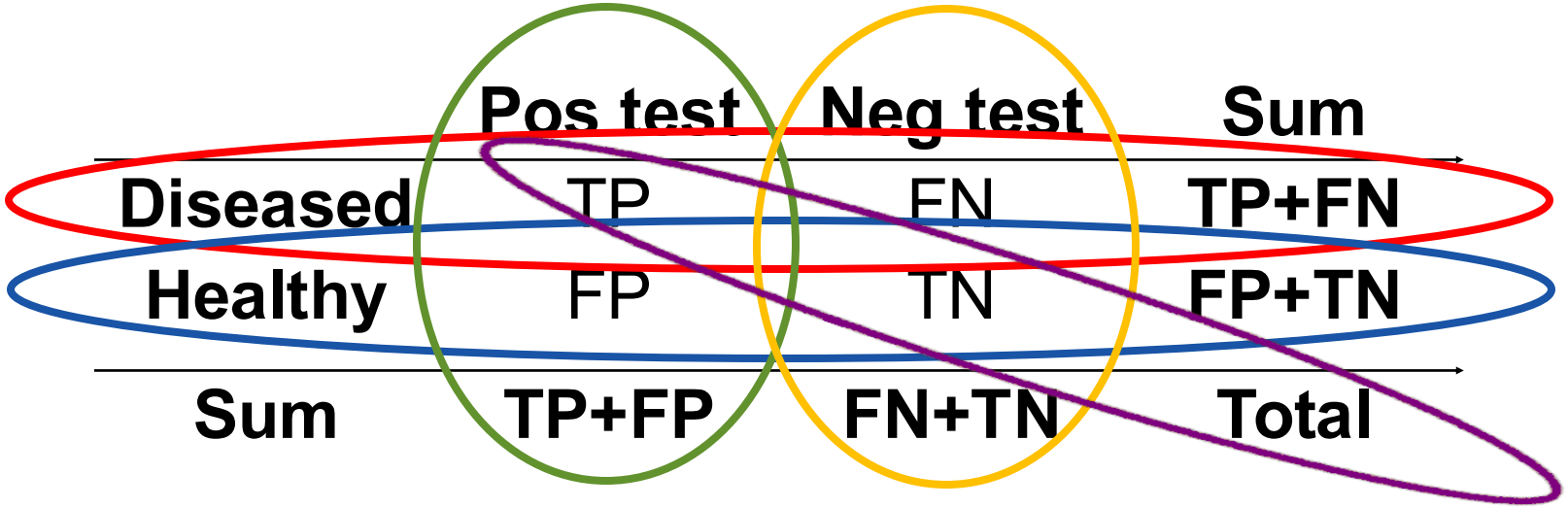
- Usually defined as the agreement with a reference method (”gold standard”, ”ground truth”)
  - Mean error: not suitable – under- and over-estimation will cancel each other out.
  - Mean absolute error, $MAE = \frac{1}{n}\sum_i |x_i - r_i|$
  - Root mean square error, $RMSE = \sqrt{\frac{1}{n}\sum_i (x_i - r_i)^2}$

    may overemphasize large individual errors
  - Maximum absolute error: $MaxAE = \max_i |x_i - r_i|$

    could easily depend on one single large error

```
error <- measured - ref
mean(abs(error))
sqrt(mean(error^2))
max(abs(error))
```

Basic concepts for a diagnostic test



- Sensitivity: $\Pr(Test = pos|Truth = pos) = TP/(TP + FN)$
- Specificity: $\Pr(Test = neg|Truth = neg) = TN/(FP + TN)$
- Pos pred value: $\Pr(Truth = pos|Test = pos) = TP/(TP + FP)$
- Neg pred value: $\Pr(Truth = neg|Test = neg) = TN/(FN + TN)$
- Accuracy: $\Pr(Test = Truth) = (TP + TN)/Total$

Confidence intervals

	Pos test	Neg test	Sum
Diseased	25	5	**30**
Healthy	15	105	**120**
Sum	**40**	**110**	**150**

- Estimate of sensitivity $\hat{p} = TP/(TP + FN) = 25/30 = 0.833$
- Exact computation of confidence interval (CI) from binomial distribution

```
binom.test(25,30)
```

- gives confidence limits of 0.653 and 0.943
- Or normal approximation (for large numbers)

$$\hat{p} \pm 1.96SE(\hat{p}) = \hat{p} \pm 1.96\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.833 \pm 1.96\sqrt{\frac{0.833 \cdot 0.167}{30}} = \begin{cases} 0.645 \\ 0.937 \end{cases}$$

```
prop.test(25,30)
```

Comparing two diagnostic tests

	Pos test A	Neg test A	Sum
Pos test B	20	5	**25**
Neg test B	15	60	**75**
Sum	**35**	**65**	**100**

- How often do test A and test B agree?
- (20+60)/100 = 80%
- ”Agreement”

Comparing two diagnostic tests

	Pos test A	Neg test A	Sum
Pos test B	20 8.75	5 16.25	**25**
Neg test B	15 26.25	60 48.75	**75**
Sum	**35**	**65**	**100**

- Actual agreement p_o= 80% — κ = 0.529
- Ideal agreement = 100% — κ = 1
- Worthless agreement p_e = 57.5% — κ = 0
  (computed from **expected frequencies** like for χ^2)

Agreement is rescaled to **kappa** values using $\kappa = \frac{p_o - p_e}{1 - p_e}$

```
library(irr)
kappa2(data)
```

Comparing two diagnostic tests

	Pos test A	Neg test A	Sum
Pos test B	20 8.75	5 16.25	**25**
Neg test B	15 26.25	60 48.75	**75**
Sum	**35**	**65**	**100**

Agreement is rescaled to **kappa** values using $\kappa = \frac{p_o - p_e}{1 - p_e}$

Using $SE(\kappa) = \sqrt{\frac{p_0(1-p_0)}{n(1-p_E)^2}}$ an approximate confidence interval for κ can be computed as $\kappa \pm 1.96\ SE(\kappa)$

```
library(irr)
kappa2(data)
```

Comparing two diagnostic tests

	Pos test A	Neg test A	Sum
Pos test B	20	5	**25**
Neg test B	15	60	**75**
Sum	**35**	**65**	**100**

- Is there a systematic difference between test A and test B?
- Consider the cases where the tests do not agree.

15 is significantly more than 5
(p = 0.04 by the binomial test).

McNemar's test

```
binom.test(c(15,5))
mcnemar.test(confusion_matrix)
```

Test differences in sensitivity and specificity

	Test A pos	Test A neg
Diseased	25	5
Healthy	15	105

	Test B pos	Test B neg
Diseased	20	10
Healthy	10	110

- Is the difference in sensitivity (0.83 vs. 0.67) significant?
- For sensitivity, look at the subpopulation of diseased subjects.
- For specificity, look at the subpopulation of healthy subjects.
- Apply the McNemar test.

Diseased	Test A pos	Test A neg
Test B pos	18	2
Test B neg	7	3

```
binom.test(c(2,7))
```

gives p=0.180

Healthy	Test A pos	Test A neg
Test B pos	9	1
Test B neg	6	104

```
binom.test(c(1,6))
```

gives p=0.125

Multiple categories

Ratings by two observers:

	Observer 2			
Obs. 1	**A**	**B**	**…**	**X**
A	n_{11}	n_{12}		n_{1k}
B	n_{21}	n_{22}		n_{2k}
…				
X	n_{k1}	n_{2k}		n_{kk}

Agreement computed from observed (p_o) and expected (p_e) frequency matrices.

kappa calculated as $\kappa = \frac{p_o - p_e}{1 - p_e}$

Using $SE(\kappa) = \sqrt{\frac{p_0(1-p_0)}{n(1-p_E)^2}}$, an approximate confidence interval for κ can be computed as $\kappa \pm 1.96\ SE(\kappa)$

The weighted Kappa

	Observer B			
Obs. A	**1**	**2**	**…**	**X**
1	n_{11}	n_{12}		n_{1k}
2	n_{21}	n_{22}		n_{2k}
…				
X	n_{k1}	n_{2k}		n_{kk}

With ordinal data, weighting can be applied to compensate for ”close misses”.

Example of weighting matrix:

Weighted agreement values are fed into the usual kappa formula.

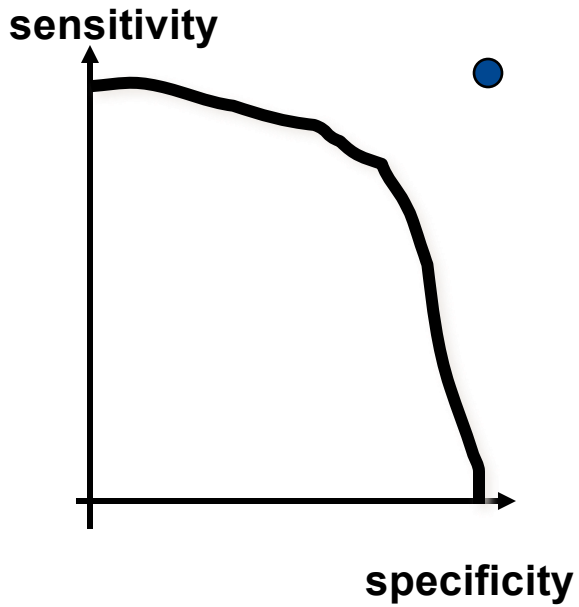
	1	2	3	4	5
1	1.00	0.75	0.50	0.25	0.00
2	0.75	1.00	0.75	0.50	0.25
3	0.50	0.75	1.00	0.75	0.50
4	0.25	0.50	0.75	1.00	0.75
5	0.00	0.25	0.50	0.75	1.00

```
library(DescTools)
CohenKappa(tableAB, weights=”Equal”)
```

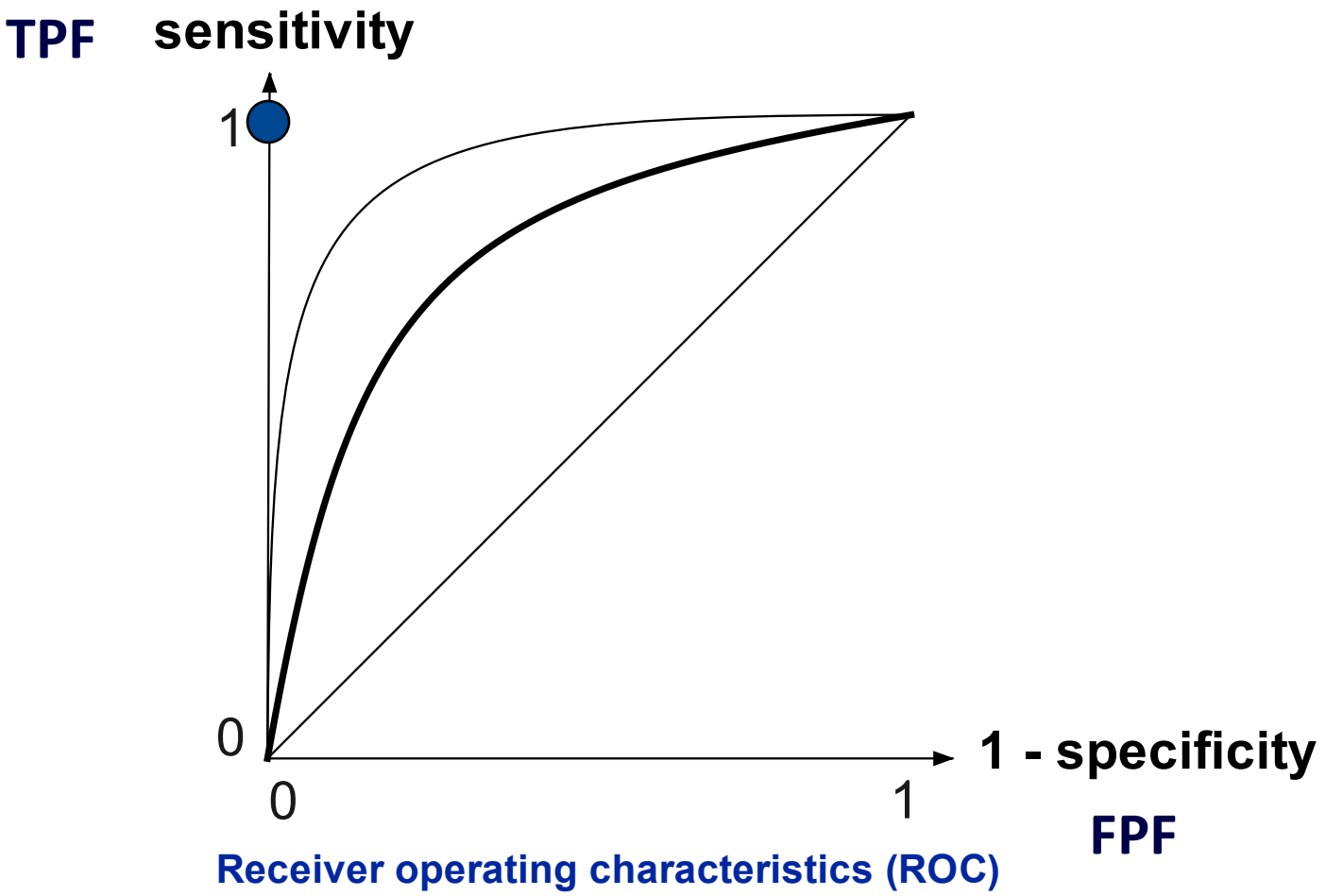
Threshold levels for diagnostic tests

- Lower threshold for pathology:
- Sensitivity increases
- Specificity decreases

- Higher threshold for pathology:
- Sensitivity decreases
- Specificity increases

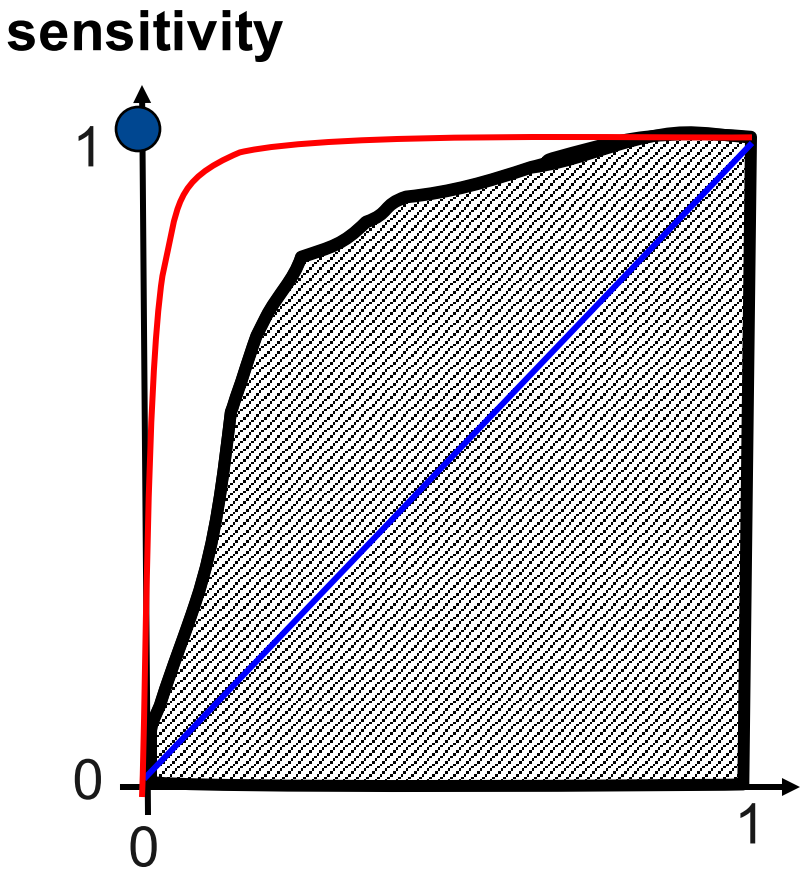


The ROC curve



Receiver operating characteristics (ROC)

The ROC curve



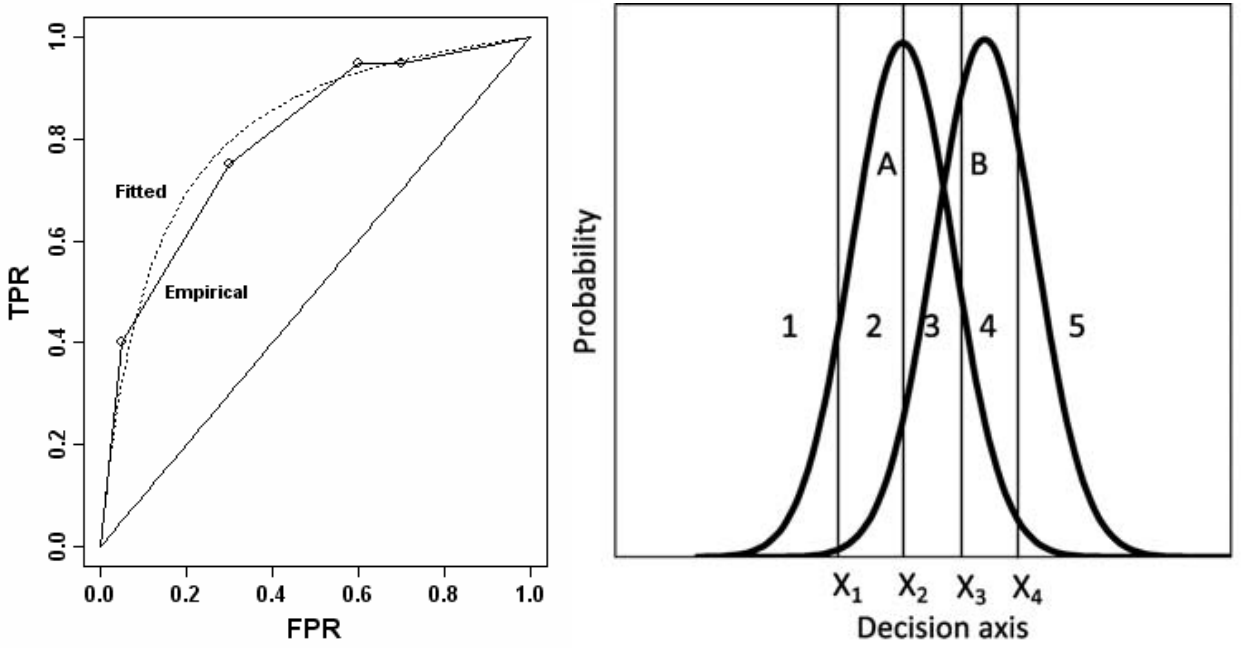
Area under ROC curve (AUROC):

1 perfect

0.5 random

The ROC curve

- Can be computed from raw data (empirical) or from a fit.
- Most common fit: assume two normally distributed populations.

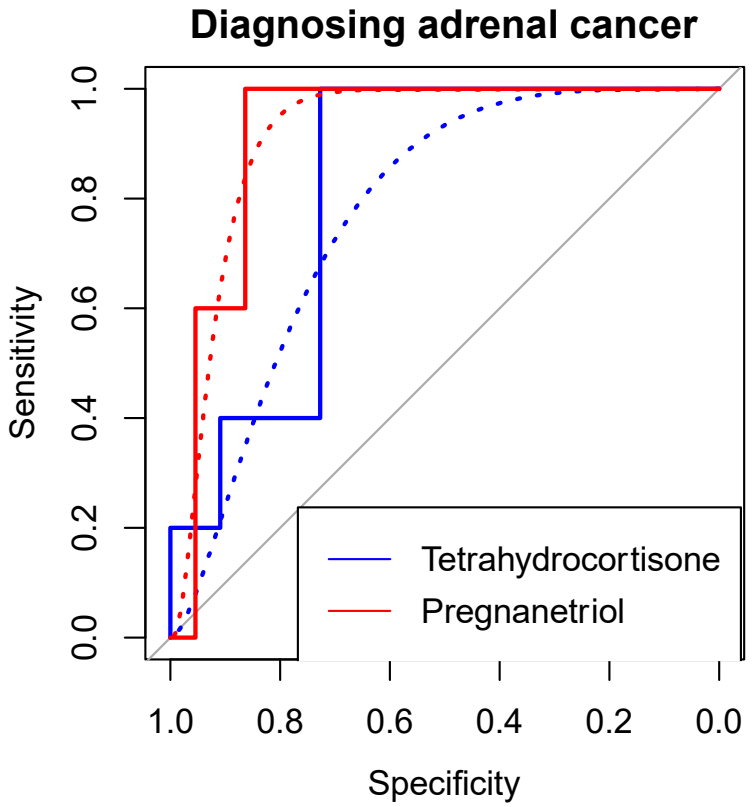


(Zhang. Int J of Biostatistics 2009; M Båth. Radiation Protection Dosimetry 2010)

ROC statistical tools

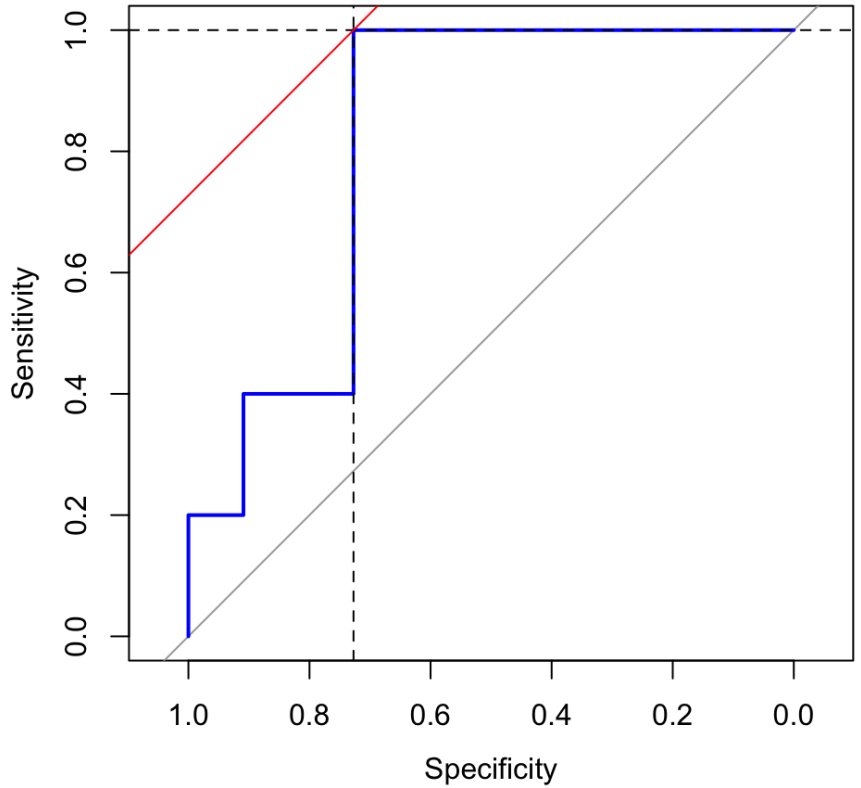
- Standard errors and confidence intervals for the AUROC can be calculated, assuming a parametric model, or by bootstrapping.
- Comparison of two ROC curves: DeLong's test.

```
roc.test(roc.th, roc.pr)
```



Finding the ”optimal” threshold using ROC

- How to find an optimal threshold separating diseased from healthy cases?
- The ROC curve gives you several choices.
- One solution: find the maximum of sensitivity + specificity.
- Depending on the application, a different linear combination than the sum may be more relevant.




```
which.max(roc.th$specificities+
roc.th$sensitivities)
```

L6-1 Statistics of Agreement (II)

Testing differences vs. studying agreement

- Difference ”proven” by a significance test.
- What can we conclude if there is no significant difference?
- The confidence interval can tell us how big the difference may be.
- How to decide whether two measurement methods give the same result? Are they exchangeable?
- Two methods to measure peak expiratory flow rate (Bland & Altman, 1986)
- Correlation is high: $r = 0.94$



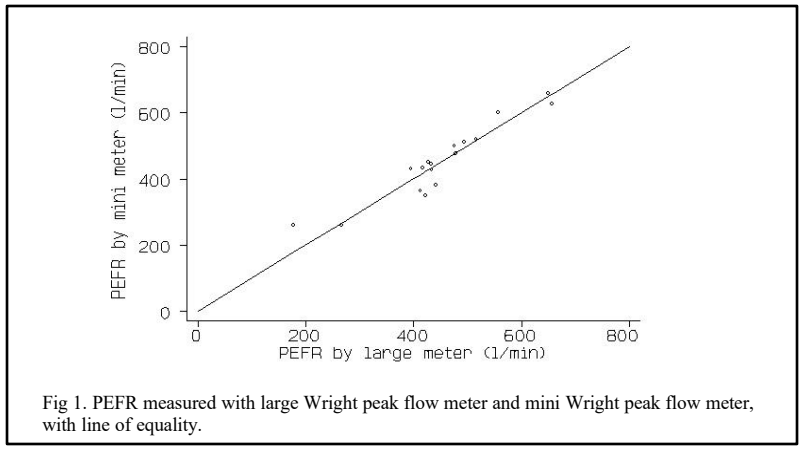


Fig 1. PEFR measured with large Wright peak flow meter and mini Wright peak flow meter, with line of equality.

The Bland-Altman plot

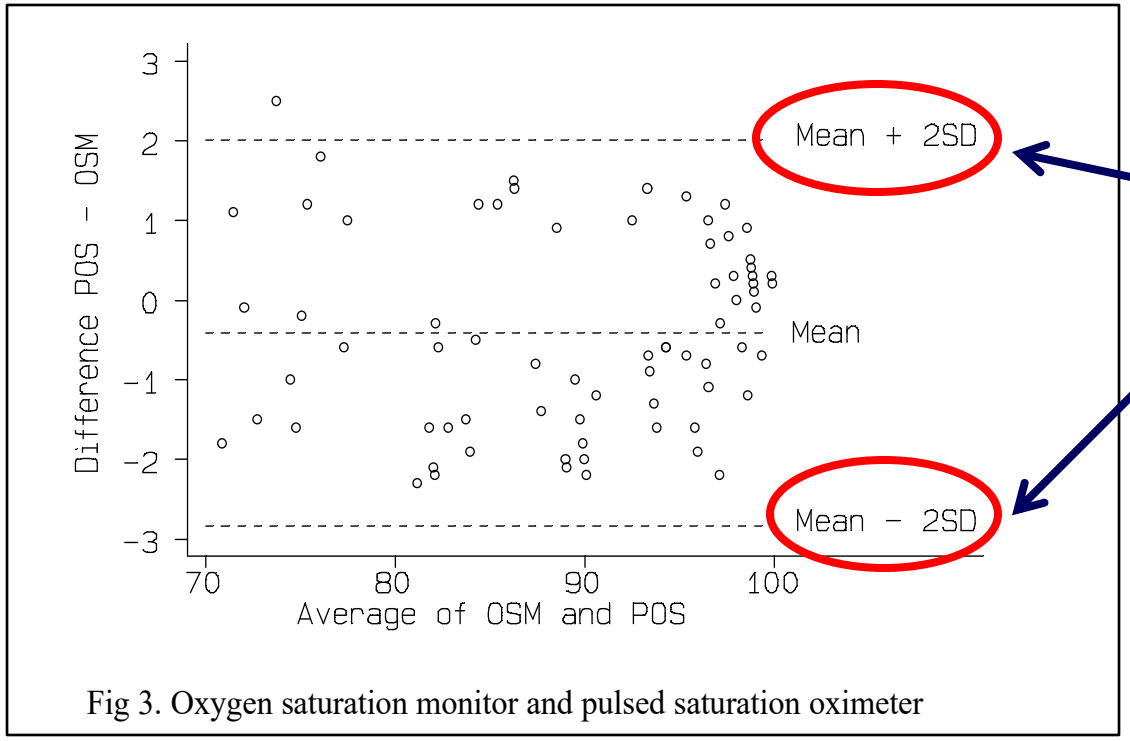


Fig 3. Oxygen saturation monitor and pulsed saturation oximeter

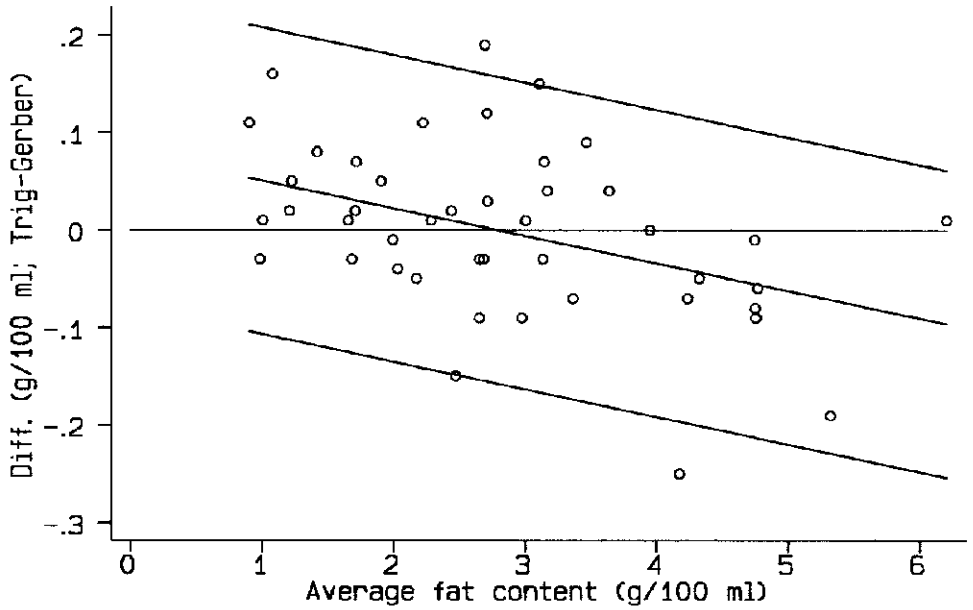
Limits of agreement

Bland & Altman 1987

Plot the difference vs. the average of the two methods.

Compute *Limits of agreement* as **mean ± 1.96·SD** of the difference.

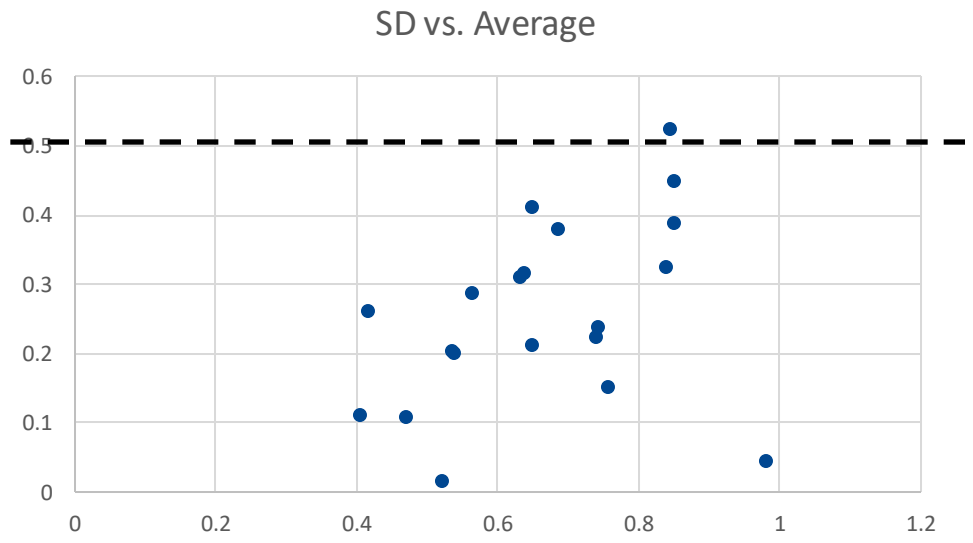
Drifting deviation



Regression-based Limits of agreement (Bland & Altman 1999)

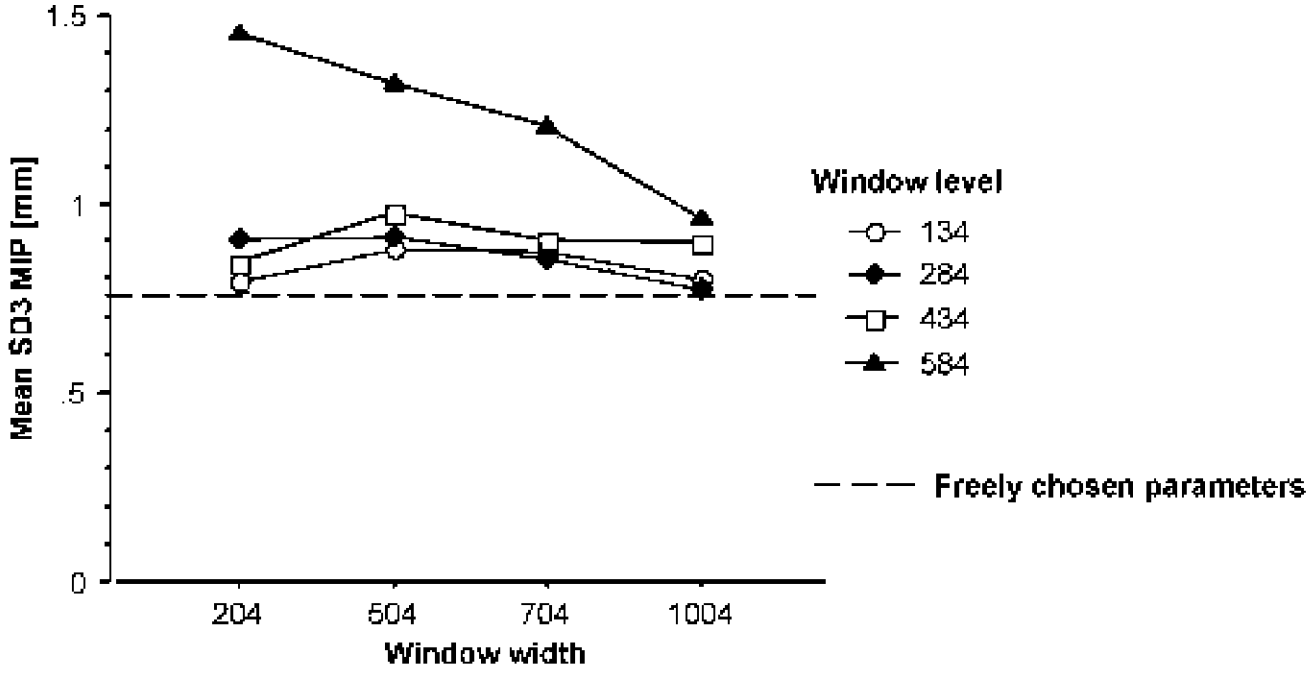
Comparing more than two methods

Plot *SD(Method1, Method2, …MethodK)* vs. average of methods.



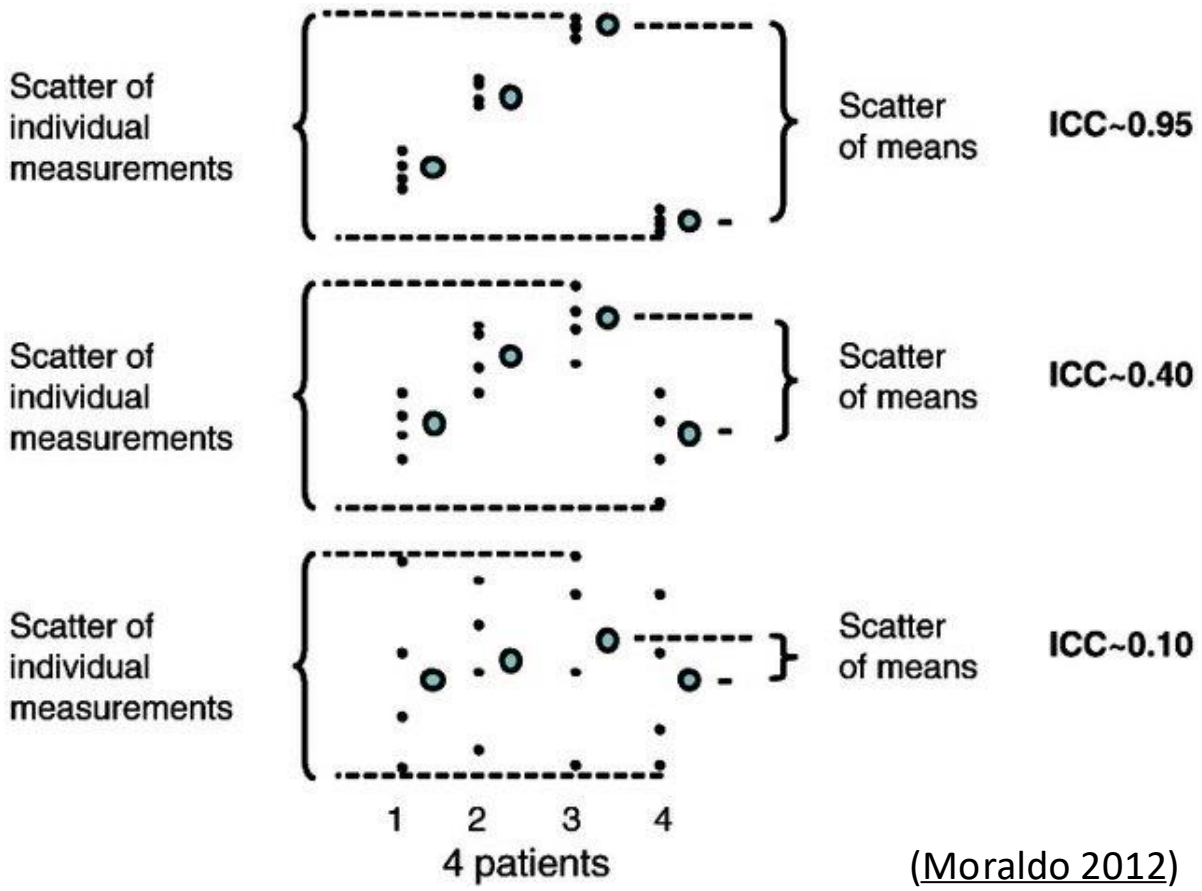
Comparing more than two observers

- Plot of SD(Observer1, Observer2, …ObserverK) vs. independent variables (categorical).
- Example: measuring the diameter of the aorta on a radiological workstation, with a number of settings, by 3 observers.



The intraclass correlation coefficient (ICC)

Compare the spread of individual measurements to the spread of means for each measurement object (subject).



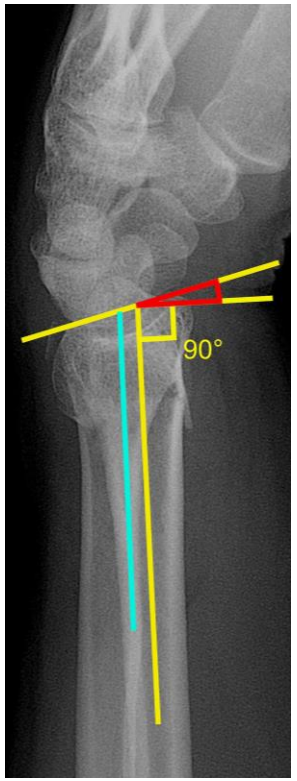
(Moraldo 2012)

The intraclass correlation coefficient (ICC)

- Compare the spread of individual measurements to the spread of means for each measurement object (subject).
- Formal approach (modern version):
- Random effects model (simplest case):
- $Y_{ij} = \mu + \alpha_j + \varepsilon_{ij}$
- $ICC = \frac{Var(\alpha)}{Var(\alpha)+Var(\varepsilon)}$
- Alt. model with systematic differences: $Y_{ij} = \mu + \alpha_j + \beta_i + \varepsilon_{ij}$

The intraclass correlation coefficient (ICC)

Example: 10 patients with radius fracture, angulation of the fragment measured by 4 radiologists (raters):



Case	Rater A	Rater B	Rater C	Rater D
1	19,5	24,2	15,3	22,7
2	18,1	14,1	12,4	11,2
3	31,0	32,1	31,8	31,6
4	16,7	15,5	13,3	20,5
5	22,7	22,8	26,9	24,8
6	49,3	44,5	43,4	46,0
7	30,6	24,5	22,0	30,7
8	14,4	15,4	17,3	19,1
9	23,5	26,6	22,0	27,4
10	11,8	13,8	19,4	15,9

The intraclass correlation coefficient (ICC)

- Concentrates agreement information into one number.
- Can be used for interval and ordinal data.
- R command **icc**() in package **rel** or **irr**.
- Choices that need to be made:
- Are all cases judged by the same raters (”twoway model”) (the most common situation) or each case by a different set of raters (”oneway model”)?
- ”Consistency” or ”agreement” between raters, i.e. should systematic errors be reflected?
- Unit of analysis: ”single” (default) or ”average”?

Ref: Shrout PE, Fleiss JL. Intraclass correlations: Uses in assessing rater reliability. Psychological Bulletin. 1979;86(2):420-8. https://dx.doi.org/10.1037/0033-2909.86.2.420

ICC vs. Random-effects linear models

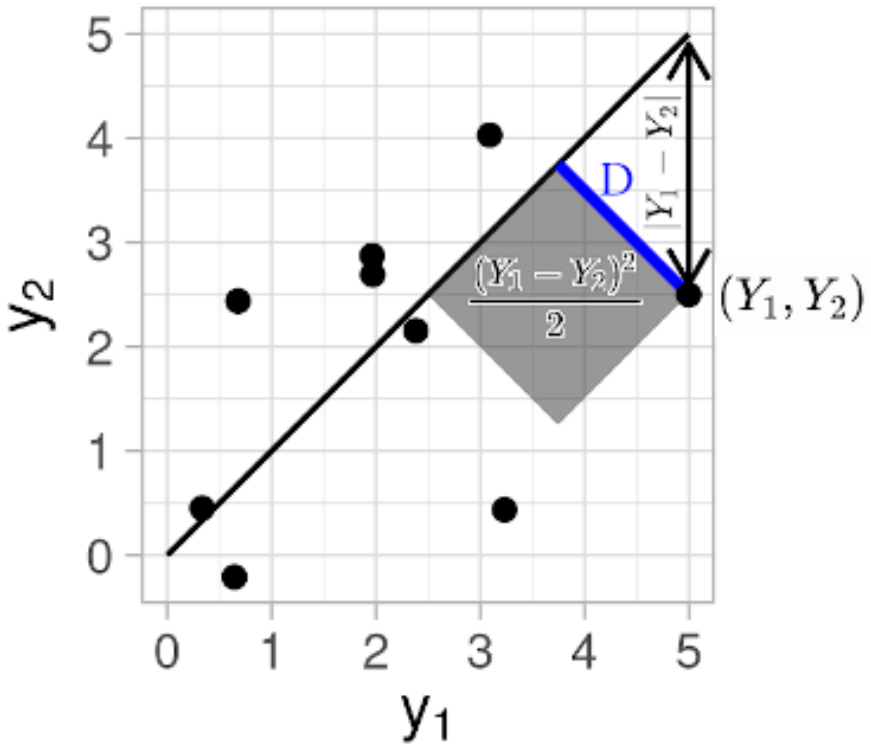
- Angles measured by 4 observers in 10 patients.
- Data in long form.
- Define a model with random effects:

```
model=lmer(angle ~ 1 + (1|Case) + (1|rater))
```

- ICC is obtained from the `summ()` command in `library(jtools)`.
- More flexible than the `icc()` command for handling more irregular designs with varying sets of raters for different cases.

Case	Rater	Angle
1	Rater_A	19,5
2	Rater_A	18,1
3	Rater_A	31,0
4	Rater_A	16,7
5	Rater_A	22,7
6	Rater_A	49,3
7	Rater_A	30,6
8	Rater_A	14,4
9	Rater_A	23,5
…	…	…
8	Rater_D	19,1
9	Rater_D	27,4
10	Rater_D	15,9

Alternatives to ICC



www.alexejgossmann.com/ccc

- Lin's concordance correlation coefficient (CCC).
- Estimate the squared distance to the identity line.
- Normalize with the sum of variances:
- $CCC = 1 - \frac{E\left((Y_1 - Y_2)^2\right)}{\sigma_1^2 + \sigma_2^2 + (\mu_1 - \mu_2)^2}$
- Numerically very close to ICC (but less general).
- Described more in Watson's paper.

Summary

- Assessing agreement may be just as important as demonstrating differences.
- Statistical tools exist for categorical data (Cohen's and Fleiss' Kappa) and continuous interval data (ICC, CCC).
- Accuracy usually means agreement with a reference method.
- For binary data, McNemar's test is useful (*e.g.*, comparing sensitivities and specificities).
- Diagnostic accuracy of continuous measurements is best evaluated with ROC analysis.
- Graphical methods are useful (Bland-Altman plots, ROC curves).